

# BOSU VANAJA

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## PROFILE

Electronics and Communication Engineering undergraduate with hands-on experience in RTL design, CMOS circuit simulation, and RTL-to-GDSII physical design flow using Cadence Virtuoso and Innovus. Interested in VLSI backend design, ASIC implementation, and digital system design.

## PROJECTS

### VLSI Physical Design Implementation (Cadence Flow)

GitHub: [github.com/bosu-vanaja/VLSI-physical-design-cadence-flow](https://github.com/bosu-vanaja/VLSI-physical-design-cadence-flow)

- Implemented RTL-to-GDSII physical design flow for digital blocks including FIFO, LFSR, FSM, register bank, counter, and 3-stage pipeline using Cadence Innovus.
- Performed floorplanning with optimized core utilization and aspect ratio
- Executed placement and routing while resolving congestion and overlap issues
- Designed Clock Tree Synthesis (CTS) and analyzed clock skew and insertion delay
- Conducted Static Timing Analysis (STA) and fixed setup/hold violations
- Performed DRC/LVS verification and parasitic/power analysis

### CMOS Circuit Design & Simulation (Cadence Virtuoso)

GitHub: [github.com/bosu-vanaja/CMOS-circuit-design-cadence-virtuoso](https://github.com/bosu-vanaja/CMOS-circuit-design-cadence-virtuoso)

- Designed and simulated CMOS circuits including CMOS inverter, NAND/NOR gates, SRAM, and differential amplifier using Cadence Virtuoso
- Created transistor-level schematics and testbench configurations
- Performed DC and transient analysis to evaluate output response and circuit behavior
- Verified functionality through waveform analysis and simulation

## EDUCATION

Bachelor of Technology in Electronics and Communication Engineering  
Dhanekula Institute of Engineering and Technology | 2023 – Present

CGPA: 9.32 / 10.0

## TECHNICAL SKILLS

|               |                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------|
| HDL           | Verilog (RTL Design, Testbench)                                                                    |
| EDA Tools     | Cadence Virtuoso, Cadence Innovus, Synopsys Tools                                                  |
| Core Concepts | Digital Electronics,RTL Design, FSM, STA, CMOS Fundamentals, CTS, Floorplanning, Routing, DRC, LVS |
| Programming   | C, C++, Python, QNX(RTOS)                                                                          |

## CERTIFICATIONS

- Advanced Physical Design Workshop – Synopsys Tools (2026)
- VLSI Design – NPTEL
- Data Structures – NPTEL
- Python Programming – NPTEL